Digital Communication

Core Data Transmission Concepts

The speed and user experience of any digital connection rely heavily on two core physical factors:

Bandwidth

- **Definition**: The number of bits that can be carried by a network connection in exactly one second.
- **Analogy**: Think of it as a water pipe; a larger diameter pipe transfers more water volume at once.
- **User Experience Impact**: Higher bandwidth accelerates file uploads/downloads and enables high-data real-time actions like multiplayer online gaming or HD video streaming.

Latency

- **Definition**: The amount of delay time it takes to send data package units between digital devices.
- **Measurement Technique**: Determined by "pinging" a web server domain name to get the round-trip return time.
- **User Experience Impact**: High latency introduces visible "lag" in online games (delayed actions) and causes a noticeable delay in live video playback feeds.

Video Streaming & Buffering

- **Buffer**: A temporary memory area holding video data so full downloads aren't required prior to playback launch.
- **Mechanism**: Continuous background downloads must fill the buffer faster than playback empties it to prevent pausing lag.

Factors Restricting Data Transfer Speed

Many environmental and physical factors disrupt or reduce available connection bandwidth:

- **Transfer Method**: Copper cable lines carry more physical wave frequencies than wireless bands, delivering higher absolute bandwidth.
- **Interference**: External electromagnetic signals from consumer appliances (microwaves, fridges) degrade signals.
- **Blockages**: Physical building walls and heavy furniture reduce wireless radio signal strength.

- **Distance**: Signal strength naturally degrades as the physical distance between data devices expands.

*Note: Cabled wires use thin metal layers called **shielding** to block electromagnetic signal interference.*

Four Major Network Classifications

When two or more computer devices connect together, they establish a network:

Network Type	Geographical Range	Primary Characteristics & Contexts
LAN (Local Area Network)	Small, restricted space (e.g., individual school, home, or office building).	High-speed local hardware data sharing.
WAN (Wide Area Network)	Large global/national space.	Links disparate corporate/government LANs; the Internet is the ultimate WAN.
PAN (Personal Area Network)	Immediately surrounding an individual person.	Syncs everyday personal accessories like a smartwatch, phone, and laptop.
Tethering	Short-range localized bridge. To share mobile data	Turns a mobile device into a host broadband modem for surrounding devices to use.

WAYS IN WHICH DIGITAL DEVICES COMMUNICATE

1. Satellite Communication

- Satellites transmit data to and receive data from digital devices.
- Digital devices use antennas to receive the radio signals that satellites transmit.
- **Benefits**: The number of satellites means that the system is always available. It also cannot be affected by power shortages.
- **Drawbacks**: Satellite signals do not pass through solid objects. This means that they will not work in areas with tall buildings or in tunnels. Signals can also be affected by atmospheric weather conditions such as heavy snow or rain.

Satellite Applications | Use

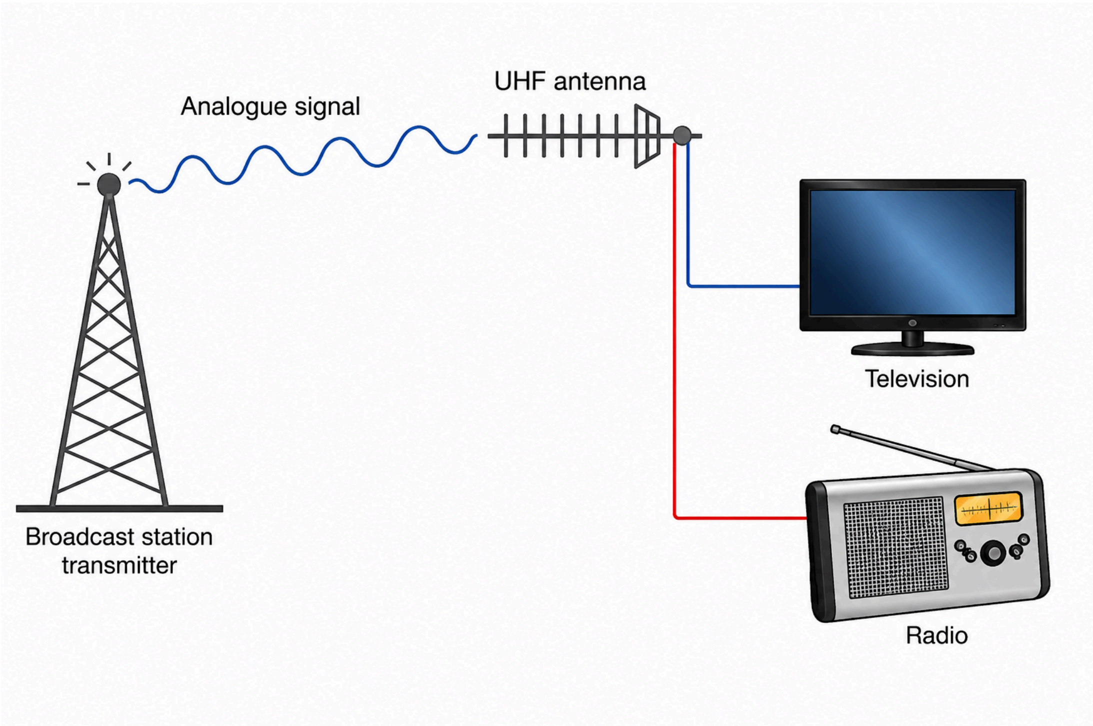
- **GPS**: Signals are sent from a network of 24 satellites orbiting the Earth. A view of only four satellites is required to calculate an accurate location.
- **Television (DVB-S)**: Digital Video Broadcasting - Satellite transmits a video signal using a large antenna on Earth to one or more satellites, which then broadcasts the

signal back down to Earth. A satellite television viewer requires an antenna and a set-top box (or an installed decoder) to decode the signal.
- **Telephone**: Used to allow people in remote areas to place voice calls using satellite telephones.
- **Military**: Used for communication systems, such as the Global Command and Control System.

2. Broadcast Communication

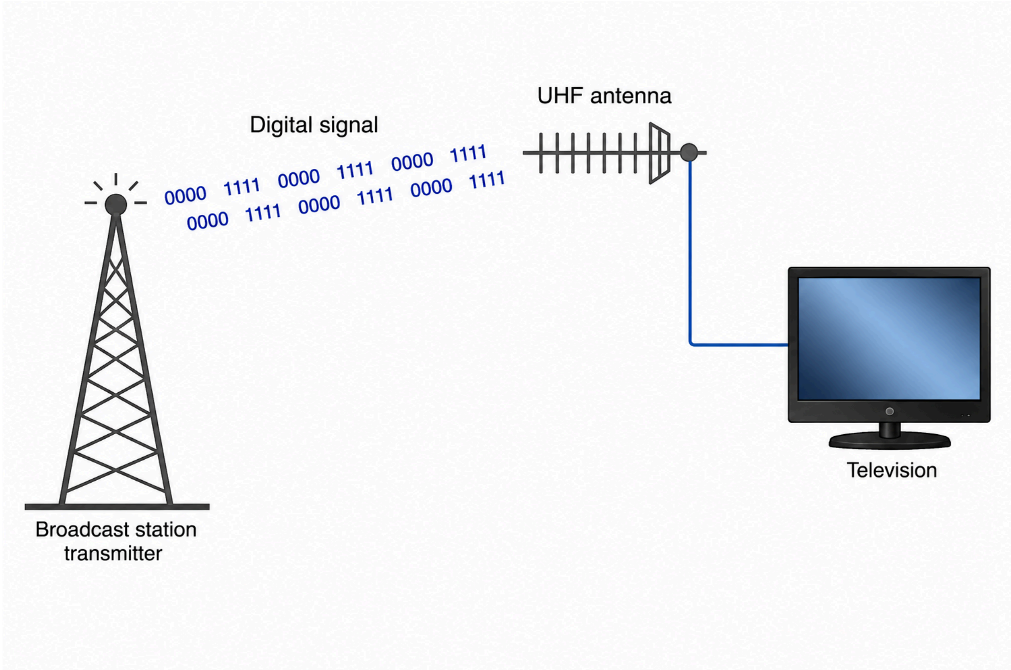
Analogue Television and Radio

- Transmitters broadcast television and radio signals that are received by a viewer's antenna.
- This antenna sends a signal through a wire to the television or radio receiver, which converts it into images and audio.



Digital Television (DVB-T)

- **DVB-T (Digital Video Broadcasting - Terrestrial)** is a method of DVB where the transmitters are based on Earth, rather than in orbit as they are in DVB-S.
- To receive digital television broadcasts transmitted by DVB-T, viewers can use the same antenna that they use to receive analogue broadcasts. They do not need a special antenna.
- **DVB-T2** is a newer standard that provides more functionality, such as High Definition Television (HDTV) and interactive services.



Digital Radio (DAB)

- **DAB (Digital Audio Broadcasting)** is broadcast in the same way as DVB.
- DAB provides more radio stations and can also carry text data (such as the time, name of the station, and details of the music being played) that DAB receivers can display.

3. Wired Communication

COMMON WIRED CONNECTION TYPES AND THEIR USES		
Connection Type	**Visual Reference**	**Primary Use Case**
HDMI Digital video connections		Transmits digital video and audio between devices, such as a computer and monitor/TV.
S/PDIF Digital audio connections		Transmits high-quality digital audio.
Minijack (3.5 mm) Personal headphones and audio		Connects personal headphones, speakers, or microphones.
USB Storage and data transfer		Transfers data and connects peripherals such as storage devices, keyboards, and mice.
Ethernet Networking connection		Connects devices to a wired network or the internet.

4. Wireless Communication Standards

Wi-Fi

- Connects local devices to an overall building access point network.
- Follows the standardized **IEEE 802.11** protocol specification rules.

Bluetooth

- Built strictly for short-range personal device connections.
- Demands explicit device **pairing** and offers less bandwidth than Wi-Fi systems.

Feature / Criteria	WI-FI	BLUETOOTH
Range	Long ✓	Short ✗
Bandwidth	High ✓	Low ✗
Power	High ✗	Low ✓
Security	High ✓	Low ✗
Can connect multiple devices simultaneously	Yes ✓	Limited ✗ (usually have to be paired)

3G & 4G (Cellular Networks)

- Often termed **mobile broadband**, managed directly by mobile phone carriers.
- Supplies internet paths when external local Wi-Fi nodes are completely out of reach.

Infra-red (IR)

- Extremely short-range.
- Requires straight line-of-sight.
- Can be blocked by solid walls and sunlight.
- Extensively chosen for simple home appliances like television remote control units.

Near-Field Communication (NFC)

- Short-proximity system using local **RFID** chips.
- Powers contactless payment bank cards, travel cards, and smartphone payment wallets.

Direct Comparison: Wired vs. Wireless

Feature	**Wired Connections (Ethernet/USB)**	**Wireless Connections (Wi-Fi/Bluetooth)**
Speed	Highly optimal, much faster.	Generally slower than cable.
Stability	Highly reliable; minimal interference.	Easily disrupted by building walls/appliances.
Security	Highly secure; requires direct physical access.	Less secure; radio signals can be intercepted.
Portability	Fixed space; constrained strictly by cable length.	Fully mobile anywhere within active signal range.
Mess & Safety	Cable clutter; physical hazard for tripping.	Exceptionally clean and completely tidy workspace.

CHAPTER QUESTIONS

1. Explain one reason why some games console controllers use Bluetooth rather than infra-red to connect to the console. (1)
2. State one way in which using infra-red in games console controllers could affect the experience of the person playing the game. (1)
3. Describe how video streaming works. (3)
4. Explain why streaming is more convenient for the user than downloading. (3)
5. State three types of wired connection. (3)
6. Which one of these best describes the internet? (1)
    i. LAN
    ii. WAN
    iii. PAN
    iv. VLAN
7. Explain why global games companies use games servers in multiple countries to ensure that the experience of users is not negatively affected when playing online games. (3)